Tool-Intent Stabilization Is Operationalization-Dependent: A Cautionary Measurement Study of Streaming Retrieval-Augmented Generation

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Abstract

Streaming Retrieval-Augmented Generation (Streaming RAG) hides tool latency by issuing retrieval queries in parallel with the user's still-arriving input, before the utterance is complete. Speculation can only help, though, when the correct query becomes determinable before the user stops speaking or typing—a property of the query, not the system. We name and measure this property, **tool-intent stabilization** (TIS): the point in the input stream at which a speculative query's retrieval converges on the answer-bearing result.

The measured value of TIS depends critically on the retrieval corpus. Against the per-question candidate pool that CRAG ships, stabilization looks early (median $\phi_{\mathrm{suf}} = 0.14$); against a realistic global corpus it is late (median up to 0.75 on NQ at $\sim$1M documents), grows later with corpus size, and holds for both sparse and dense retrieval (FiQA: BM25 0.636, dense 0.625).

This corpus-dependence has a direct consequence for the hideable-latency headroom, and the gap grows with tool latency. At a large tool latency (L=2500 ms, the regime streaming RAG targets) the streamable fraction falls from 39.2% (per-question corpus) to 9.2% (global NQ)—whereas at small tool latency (L=600 ms) long queries still capture meaningful headroom in both conditions.

The TIS framework (t_{sc}, t_{suf}, ϕ, and the bound H) is sound; the caution is to measure TIS against the corpus the deployed system retrieves over. On the CRAG benchmark (1371 validation questions) we (i) characterize how stabilization is distributed; (ii) derive a model-agnostic bound H on the latency hideable behind the remaining input; (iii) validate H against a working pipeline; and (iv) identify query properties that predict early versus late stabilization. The study needs no model training and runs on commodity CPU hardware.

1 Introduction

Tool-augmented dialogue systems increasingly rely on external retrieval to remain factual, but tool calls add latency that disrupts conversational flow. Streaming RAG (Arora et al., 2025) addresses this by predicting and issuing tool queries in parallel with the user's still-arriving input, then reflecting on whether retrieved results suffice. The approach reports sizable accuracy and latency improvements and is modality-agnostic, applying equally to typed input.

The reported latency benefit, however, is an aggregate over a benchmark. It leaves a basic question unanswered: **for which queries can speculation help at all, and by how much?** The answer is not a property of the model, the retriever, or the speech stack. It is a property of where, within an utterance, the information that determines the tool query first appears. If the decisive term arrives last ("who makes the console called the *PlayStation*"), no early speculative query can retrieve the right evidence, and the latency win collapses to zero regardless of system quality. If the decisive term arrives early, almost the entire tool latency can be hidden behind the remaining input.

We initially measured tool-intent stabilization against the per-question candidate pool that CRAG ships with each query, and that measurement made stabilization look early (median $\phi_{\mathrm{suf}} = 0.14$). Measuring instead against a realistic global corpus—the corpus the deployed system actually retrieves over—reveals that early picture to be largely an artifact of the per-question pool: stabilization is in fact *late* (median ϕ_{suf} up to 0.75 on NQ at $\sim$1M documents), grows later with corpus size, and holds for both sparse and dense retrieval. This corpus-dependence is the finding this paper is organized around.

We study this directly. By characterizing tool-intent stabilization as a measurable, query-intrinsic quantity, we obtain: (i) a model-agnostic *aggregate* bound H on the share of tool latency that can be hidden behind ongoing input, computable *before* any system is built, but *operationalization-sensitive*—the corpus the bound is evaluated on determines how optimistic or conservative it is; (ii) a measurement caution backed by global-corpus evidence—dose-response across corpus sizes, two benchmarks (NQ and FiQA), and two retriever families (BM25 and dense)—showing that per-question pools substantially over-estimate real-world headroom, and that headroom shrinks as tool latency grows; (iii) a principled account of the failure mode (late stabilization) and a coarse build/skip signal for whether a learned trigger is worth its compute cost; and (iv) results that transfer across text and speech because the quantity is modality-agnostic. Our contribution is measurement and analysis rather than a new system.

2 Background and Related Work

Retrieval-Augmented Generation (Lewis et al., 2020) grounds generation in retrieved evidence; dense passage retrieval (Karpukhin et al., 2020) and the sparse BM25 model (Robertson and Zaragoza, 2009) are the standard retrieval backbones. Streaming RAG (Arora et al., 2025) adds a speculative dimension: a *Trigger* decides when to fire a tool query from a partial utterance, multiple speculative *Threads* run in parallel, and a *Reflector* judges whether results suffice. The idea is closely related in spirit to speculative decoding (Leviathan et al., 2023), which hides latency by doing work that may be discarded; here the discarded work is a premature tool call. Our analysis is orthogonal to the system: we quantify the headroom that any such system can exploit, set by the query alone. Two literatures tell us why such headroom should exist and how to measure when a prefix is safe to act on.

2.1 Incremental comprehension and anticipation

Why expect tool intent to settle before a query is complete? Human comprehension already works this way. That the referential intent of an utterance is fixed incrementally—often well before the utterance ends—is among the most robust findings in psycholinguistics. Altmann and Kamide (1999) showed that a verb such as "eat" directs anticipatory attention to edible objects before they are named, so partial input already restricts what can follow. Kamide et al. (2003) extend this cross-linguistically: in head-final Japanese, case-marked pre-verbal arguments let comprehenders anticipate a forthcoming argument even before the verb (the head) is heard—*pre-head* anticipation (Kamide and Mitchell, 1999). This is the human analogue of firing a tool query from a prefix: *where* the disambiguating cue sits—varying with word order and case morphology—governs how early intent stabilizes.

2.2 Stabilization over a growing prefix

If comprehension fixes intent early, an engineered system faces the dual problem: deciding when a prediction over a growing input prefix becomes safe to commit. This is the central question of *simultaneous* and *incremental* NLP. Simultaneous machine translation must decide, token by token, whether enough source has arrived to emit output: wait-k and prefix-to-prefix policies (Ma et al., 2019) and learned read/write agents (Gu et al., 2017) trade latency against the risk of committing before the disambiguating word arrives—precisely our L-versus-residual tension. Grissom II et al. (2014) cast the same problem as an agent that waits until it can confidently predict the sentence-final verb before committing; its failure mode, premature commitment forcing revision, is exactly what our volatility V records. Closest to our measurement is incremental spoken language understanding: DeVault et al. (2011) build a meaning frame from partial ASR and quantify understanding as a function of prefix length, and the incremental-unit framework (Schlangen and Skantze, 2011) formalizes the same prefix-monotonicity and revision concerns. These efforts target a *closed* intent or dialogue-act ontology. Our contribution is to port the lens to *tool/retrieval* intent over an open-world knowledge benchmark: rather than learning a commit policy, we *measure* the query-intrinsic point at which the retrieval target stabilizes, which upper-bounds what any such policy could hide.

3 Problem Formalization

Let a query Q be a word sequence $q_{1:n}$ of length n words, and let $q_{1:t}$ denote the prefix of the first t words (we measure stabilization in words throughout, matching the whitespace tokenization used by both the retriever and the input-cadence model below). Let R be a retriever returning a ranked document list; write $d_t = \text{top-1}(R(q_{1:t}))$ for the top document retrieved from the prefix, d_n for the full-query top document, and $d^\star$ for the gold answer-bearing document.

Self-consistency stabilization t_{sc}. the smallest t such that for all $s \in \{t, \ldots, n\}$, $d_s = d_n$: every prefix from t onward retrieves the same top document as the full query.

Sufficiency stabilization $t_{\mathrm{suf}}(q; C)$. the smallest t such that $d^\star \in$ top-$k(R(q_{1:t}; C))$: retrieval over corpus C already surfaces the gold evidence in the prefix. This is the operationally meaningful notion, since the full query itself may retrieve imperfectly. The corpus C is an explicit parameter: our v1 results are $t_{\mathrm{suf}}(\cdot;\, C_{\mathrm{perq}})$, where C_{perq} is CRAG's per-question candidate pool; the global-corpus arm reports $t_{\mathrm{suf}}(\cdot;\, C_{\mathrm{global}})$ over a realistic open-domain index. Because C_{perq} is far smaller and topically filtered, $t_{\mathrm{suf}}(\cdot;\, C_{\mathrm{perq}})$ systematically under-estimates how late stabilization actually occurs in deployment; all sufficiency results in this paper specify which corpus is in use.

Stabilization fraction $\phi = t^\star/n \in (0, 1]$, reported for both t_{sc} and t_{suf}. Lower ϕ means earlier stabilization and more headroom to hide latency.

Stabilization volatility V, the number of top-1 changes occurring *after* d_n is first reached. High V signals early-commit risk: a confident-looking early result that a later token would overturn.

Hidden latency.

$$H(Q; L, \delta) = \min\Big(L,\ \max\big(0,\ \tfrac{n-t^\star}{\delta}\big)\Big), \qquad (1)$$

where L is tool latency (ms) and δ is the input cadence in words per second, so that $(n - t^\star)/\delta$ is the residual input time (in seconds, scaled to ms) still to arrive after the query has stabilized. H is the portion of tool latency that completes behind the user's remaining input. It bounds the *input-hideable* component of the per-query perceived-latency saving—not the total saving, since a real pipeline can additionally overlap other costs (§5.4). A slower cadence (smaller δ) lengthens the residual and so raises H when tool latency is the larger term.

Streamable. $\mathrm{Streamable}(Q; L, \delta, \theta)$ holds iff $H \geq \theta L$ for a coverage threshold θ (e.g. $\theta = 0.8$ hides at least 80% of tool latency). The streamable fraction of a workload is the central deployment-relevant statistic.

Research questions. We address four pre-registered questions. **RQ1** asks how ϕ is distributed across queries; under the per-question operationalization (C_{perq}) the distribution is right-skewed with an early-stabilizing mass and a thin late tail—and a central question is whether that early mass is intrinsic to the queries or an artifact of the small, topically-filtered corpus, which the global-corpus arm answers. **RQ2** asks what fraction of queries admit latency hiding as L, δ, and θ vary; we expect that for large L the binding constraint is the residual input time $(n - t^\star)/\delta$, so a *slower* cadence paradoxically *raises* the streamable fraction. **RQ3** asks whether the H bound matches a working pipeline. **RQ4** asks which query features predict early vs. late stabilization; we expect question type to be the dominant signal, with simple lookups stabilizing earlier than multi-hop or aggregation queries. We address RQ1–RQ4 in §5 and include top-k and dense-retriever robustness arms (Table 3, §5.6) as sensitivity checks. A cross-linguistic condition is left to future work. We report disconfirmations alongside confirmations: RQ2 is confirmed, but the RQ1 distribution is right-skewed rather than bimodal (§5.1), and the RQ4 ordering is governed by entity position rather than reasoning complexity (§5.2).

4 Method

Data. We use CRAG, the Comprehensive RAG Benchmark (Yang et al., 2024), Task 1&2 dev set: 2706 questions, of which 1371 are the validation split we analyze. Each question carries up to five retrieved web pages, a gold answer (plus alternates), and a question-type label.

Grounding $d^\star$. CRAG provides no gold-*passage* label, only gold answer strings. We derive $d^\star$ by grounding the normalized gold answer(s) into passage text: substring match for multi-token answers, and word-boundary match for single-token answers, over the answer and its alternates. Questions whose answer is never a literal span (false-premise, "I don't know", and many aggregation/dynamic answers) are *ungroundable* and excluded from t_{suf}; we report the groundable rate explicitly. Because t_{sc} requires no grounding, we report it as a grounding-free robustness check alongside t_{suf}.

String grounding is a recall-oriented heuristic and can produce false positives. A single-token gold answer (a year, a number, "yes"/"no") may match a passage coincidentally, registering sufficiency before the genuinely answer-bearing evidence arrives. This biases t_{suf}, and thus ϕ_{suf}, early. Multi-token answers, which require a contiguous substring match, are far less prone to this. We therefore read ϕ_{suf} as a lower bound on stabilization time on the groundable subset, and treat the per-question early-stabilization finding as directionally robust rather than exact within C_{perq}.

To bound the population-selection bias this strictness induces, we also run a *relaxed* grounding arm (§5.6): for multi-token answers it accepts a passage if all of the answer's content tokens (stopwords dropped) appear anywhere in it—bag-of-content-token containment, dropping only the contiguity requirement. This is deliberately higher-recall and lower-precision than the substring rule, so the two arms *bracket* the true groundable population and the ϕ_{suf} that goes with it.

Passages and retrieval. Page HTML is cleaned to text and chunked into 120-word windows with a 20-word overlap (stride 100), capped at 400 chunks per question. We index each question's passages with a pure-Python lexical BM25 (Robertson and Zaragoza, 2009) (k_1=1.5, b=0.75) and retrieve over every prefix $q_{1:t}$, $t = 1 \ldots n$, recording d_t and the top-k set at each prefix. We sweep $k \in \{1, 3, 5\}$ for the sufficiency definition. All retrieval is deterministic; the analysis uses the CRAG `dev_v4` snapshot (the released web pages, so retrieval is fixed and not subject to live drift). The two stochastic or selection steps are seeded and reproducible: the 60-question RQ3 subset is simply the first 60 retrieved-gold questions in dataset order (no sampling), and the bootstrap confidence intervals (§5.2) use 10,000 resamples at seed 0.

Latency model and harness (RQ3). We stream each query as a uniform word sequence—a controlled proxy for input cadence δ—and sweep (L, δ, θ) analytically over the grid $L \in \{100, 300, 600, 1000\}$ ms, $\delta \in \{2, 3, 4\}$ w/s, $\theta \in \{0.5, 0.8, 1.0\}$ to obtain the streamable surface (RQ2). For RQ3 we replay a subset of questions through a faithful asynchronous Streaming RAG harness (fixed-interval Trigger, parallel Threads, Reflector) calibrated with the per-stage latencies reported by Arora et al. (2025) (Table 3): query generation $\approx$ 590 ms and fuse/reflect $\approx$ 2520 ms, in addition to the swept tool latency L. We compare the *measured* perceived-latency saving (baseline minus streaming) against the H bound.

Reproducibility. The pipeline is training-free and CPU-only. Code and the derived $d^\star$ passage labels (the only non-trivial artifact, since CRAG ships none) are available at `https://github.com/elroy-galbraith/stablize_CRAG`; every per-question metric and the full (L, δ, θ) grid can be recomputed offline without re-running retrieval.

4.1 Global-Corpus Measurement Condition

The per-question results above use C_{perq}, each question's own candidate pool. To test whether the observed early-ϕ_{suf} behaviour survives a realistic open-domain index, we add a second measurement condition that indexes all queries against a *shared*, much larger corpus. We implement this in `experiments/global_corpus.py`.

Benchmarks. We use BEIR-NQ (Kwiatkowski et al., 2019) and BEIR-FiQA (Maia et al., 2018), loaded from the BEIR HuggingFace repositories. Both benchmarks supply explicit gold qrels (relevance judgements), so $d^\star$ is given directly rather than derived by string matching. This removes the string-grounding caveat that affects the per-question CRAG arm: every query has a reliably labelled gold passage, and the groundable rate is therefore 100% by construction.

Gold-guaranteed subsampling. To allow controlled dose-response experiments at varying corpus sizes without sacrificing recall, we use a *gold-guaranteed* corpus construction: for a given query, the corpus is the union of *all* gold passages (union over all qrels entries) together with N reservoir-sampled distractor passages drawn from the remaining non-gold corpus. Every query's gold is therefore guaranteed to be present regardless of N. We sweep N across values from 10^3 to $\sim 10^6$ (the full corpus); the full-corpus data points represent the large-N limit of this dose-response. Distractors are reservoir-sampled in a single streaming pass over the corpus using a seeded RNG (seed 0), so results are deterministic and reproducible without storing the full index.

Retrievers. We evaluate two retrievers. The *sparse* arm uses `bm25s` (Lassance and Clinchant, 2024), a Lucene-style BM25 implementation with k_1=1.5, b=0.75; these are the same hyperparameters as the per-question arm, though the underlying engine differs (pure-Python BM25 vs. `bm25s`).

The *dense* arm uses `all-MiniLM-L6-v2` (sentence-transformers (Reimers and Gurevych, 2019)), encoding passages once at index time and ranking query prefixes by cosine similarity. Both retrievers follow the same prefix protocol as the per-question arm: t_{suf} is the smallest t such that some gold passage $d^\star$ appears in the top-k list at prefix $q_{1:t}$, and $\phi_{\mathrm{suf}} = t^\star/n$ is the corresponding stabilization fraction.

5 Results

We analyze the 1371 validation questions of CRAG Task 1&2. Under our string-grounding procedure, 35.4% of questions are *groundable* (the gold answer appears verbatim in the retrieved pages); the remainder—dominated by false-premise items and answers that never occur as a literal span (many aggregation and dynamic questions)—are excluded from sufficiency statistics. At k=3, BM25 surfaces the gold passage in the top-k at some prefix for 21.3% of all questions. All reported ϕ_{suf} statistics are conditional on this retrieved-gold population; ϕ_{sc} is defined for every question and serves as a grounding-free check.

This retrieved-gold subset is not a random sample of the benchmark: it is the intersection of questions whose answer is verbatim-groundable *and* whose gold passage BM25 can surface from some prefix. Both filters plausibly select for queries with a salient, early lexical anchor, so the conditional ϕ_{suf} statistics should be read as characterizing this favorable slice, not "most answerable questions"; if anything, the selection biases the reported stabilization *earlier* than it would be on the full workload. We therefore lean on ϕ_{sc}, defined on all 1371 questions, as the grounding-free companion to every sufficiency claim, and we bound the magnitude of the selection effect with a relaxed-grounding arm below (§5.6): recovering a sixth more of the population moves ϕ_{suf} by under 0.02, so the early-stabilization *pattern within* C_{perq} is robust to grounding relaxation—though as the global results show (§5.1), that per-question pattern is corpus-dependent rather than deployment-general.

5.1 Stabilization depends on the retrieval corpus

Table 1 is the headline result of this paper. Three confirmations reinforce one another.

(1) Dose-response. On NQ, median ϕ_{suf} rises monotonically with corpus size: 0.571 at gold+1k, 0.625 at gold+10k, and 0.75 at the full ~1M-document index. The pattern is exactly what the corpus-dependence account predicts: as the distractor pool grows, the retriever must be more precise about the query to surface the gold passage, and precision requires more of the utterance.

Table 1: Dose-response of ϕ_{suf} across operationalizations. Median ϕ_{suf} rises monotonically with corpus size on NQ and converges to a similar late value on FiQA. The t_{suf}=1 share collapses from ≈50% per-question to ≤3% global, exposing the per-question value as a measurement artifact.

Operationalization	median ϕ_{suf}	t_{suf}=1
CRAG per-question, string-grounded	0.14	49%
CRAG per-question, precisely-grounded	0.35	14%
Global NQ gold+1k	0.571	—
Global NQ gold+10k	0.625	—
Global NQ ~1M	0.75	0.7%
Global FiQA 10k BM25	0.636	2.8%
Global FiQA 10k dense	0.625	—

(2) Two benchmarks. The late-stabilization finding is not confined to NQ. On FiQA at gold+10k—a financially-focused QA benchmark with different topical coverage and query style—BM25 reaches median $\phi_{\mathrm{suf}} = 0.636$, in close agreement with the NQ figure at the same corpus size (0.625). Late stabilization holds across benchmarks.

(3) Retriever-general. On the identical FiQA gold+10k corpus, a dense bi-encoder (`all-MiniLM-L6-v2`) yields median $\phi_{\mathrm{suf}} = 0.625 \approx 0.636$ (BM25). The dense retriever surfaces gold for *more* queries (481 vs. 364 groundable questions): it is a better retriever. But it reaches gold at the same late stabilization point. Dense retrieval thus *generalizes* the artifact rather than rescuing the early claim.

The t_{suf}=1 collapse. Under realistic global retrieval, one token is almost never sufficient: the t_{suf}=1 share falls from ≈50% per-question (49%) to ≤3% globally (0.7% on NQ, 2.8% on FiQA), confirming that the large per-question mass at t_{suf}=1 is a corpus-size artifact, not a signal of genuine one-word disambiguation.

Intermediate control. The `global_corpus.py` top-100 `perq` arm—which pools the top-100 retrieved candidates across all queries before indexing—is an intermediate control: it yields later stabilization than CRAG's native per-question pool but earlier than the full-global arm. It is not the headline contrast; we include it as a dose-response step between C_{perq} and C_{global} to confirm the monotone size effect.

Per-question view (C_{perq}). The remainder of this section reports results under the *per-question* operationalization C_{perq}: each question is indexed against its own small retrieved-passage pool, as in the standard CRAG evaluation setup. The global results above show that C_{perq} systematically overestimates early stabilization relative to a realistic corpus. We retain this analysis for two reasons: it is the standard CRAG setup that prior work uses, enabling direct comparison; and RQ3 validates the hidden-latency bound mechanism within it. Numbers in §§5.2–5.6 should be read as characterizing *the optimistic operationalization*, not the deployment-realistic one.

5.2 RQ1: How is stabilization distributed (C_{perq})?

Under the per-question pool C_{perq}, the two stabilization notions behave very differently (Fig. 1). Self-consistency stabilizes *late*: the lexical top-1 keeps changing until, on average, ϕ_{sc} reaches mean 0.67 (median 0.73)— roughly three-quarters of the way through the query. Sufficiency, in contrast, stabilizes *earlier* within C_{perq}: the gold passage enters the top-k at mean $\phi_{\mathrm{suf}} = 0.26$ and median 0.14. This point estimate is sensitive to grounding *precision*, however—short or ubiquitous gold answers match many passages and pull it low—and rises to median 0.35 (mean 0.44) on specifically-grounded questions (§5.6). We therefore read ϕ_{suf} as a range whose median spans $[0.14, 0.35]$: early-to-moderate rather than uniformly early, but in every case settling well before self-consistency does. The gap is the central empirical observation of this study: **the answer-bearing document is retrievable from a short prefix long before the query's lexical top-1 settles**. For streaming tool use this is the favorable regime—one need not wait for the utterance to "finish" for the right evidence to be in hand.

Within C_{perq}, the ϕ_{suf} distribution (Fig. 1) is right-skewed with a large mass near zero and a thinner late tail, rather than cleanly bimodal as H1 anticipated (the global results in §5.1 show the zero-

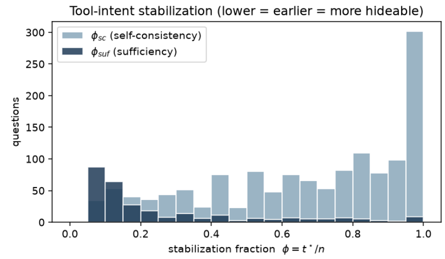

Figure 1: Distribution of stabilization fraction ϕ at k=3. Sufficiency (ϕ_{suf}, dark) concentrates near zero—gold evidence is retrievable early—while self-consistency (ϕ_{sc}, light) is late and broad.

mass largely collapses under a realistic corpus). Volatility is modest: the post-stabilization top-1 changes at all for only 20.9% of questions (mean $V = 0.52$), so early-commit risk is concentrated in a minority of queries rather than pervasive.

5.3 RQ4: What predicts early vs. late stabilization (C_{perq})?

Within the per-question pool C_{perq}, question type produces a coarse early/late split in ϕ_{suf}, spanning a roughly threefold range across medians (Fig. 2). A Kruskal–Wallis test across all eight types rejects equality of the ϕ_{suf} distributions (H=17.0, df=7, p=0.017); restricting to the five classes with $n \geq 10$—which carry the effect—the test is stronger (H=13.9, df=4, p=0.008). The effect, though significant, is *small*: the rank-based effect size is ε^2=0.06 across all eight types and 0.05 for the five large classes, i.e. question type explains on the order of 5–6% of the rank variance in ϕ_{suf}. Consistent with a small effect, the interquartile ranges in Fig. 2 overlap substantially for adjacent types: the ordering is reliable only at the extremes (*aggregation* and *comparison* earliest, *set* latest—`set` confirmed at n=20 under the relaxed-grounding arm, §5.6), not as a strict cell-by-cell ranking, and `post-processing` (n=5) has too few points to place reliably. A Dunn's post-hoc with Holm correction makes this precise: *no* pairwise contrast survives at α=0.05 (the closest, `simple` vs. `aggregation`, reaches p_{Holm}=0.05). The omnibus effect is therefore carried by the overall rank distribution, not by any single separable pair. We accordingly claim a significant but coarse type-level effect—a robust early/late *tendency*, not a precise per-type order. The qualitative ordering refines hypothesis H3 rather than confirming it. H3 expected simple lookups to sta-

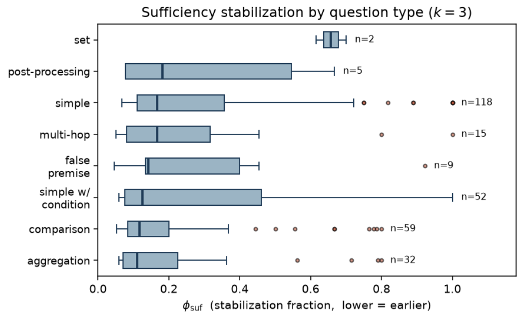

Figure 2: Distribution of ϕ_{suf} by question type (k=3, sorted by median). The 3× range across medians is real, but box overlaps are large for adjacent types—only the extremes (*aggregation*/*comparison* earliest, *set* latest) are reliably ordered. Ordering tracks entity position, not reasoning complexity.

bilize early and aggregation/comparison/multi-hop to stabilize late. We instead find *aggregation* and *comparison* among the *earliest*-stabilizing types, with *set* questions latest. The explanation is that sufficiency measures when the gold *document* becomes retrievable, not when the answer can be *computed*: a comparison or aggregation question often names its key entities up front, so the answer-bearing page is retrievable early even though producing the final answer requires post-retrieval reasoning. In other words, retrieval-sufficiency stabilization is governed by entity position in the utterance, which is not aligned with reasoning complexity. This entity-position account is carried by the well-populated classes (`simple`, `comparison`, `aggregation`, `simple_w_condition`), so it does not hinge on the noisy small-n cells. We present it as an explanation motivated by the data rather than one we have isolated: it yields a directly falsifiable prediction—that the word position of the first named entity should track t_{suf} more tightly than question type does—which a lightweight entity-tagging pass over the queries could test without any new retrieval. If that correlation held, first-entity position would also furnish a far cheaper deployment signal than the full prefix-retrieval sweep. We flag both as the natural next measurement.

5.4 RQ2: What fraction of queries admit latency hiding?

Using $t^\star = t_{\text{suf}}$ where available and falling back to t_{sc} otherwise, we compute the streamable fraction over the full (L, δ, θ) grid (Fig. 3). At the central operating point (L=600 ms, δ=3 w/s, θ=0.8), 73.9% of queries admit hiding at least 80% of tool latency.

This blended figure mixes two populations and two notions of stabilization, and we report it decomposed. On the 21.3% of queries with retrieved gold—where $t^\star = t_{\text{suf}}$ reflects evidence actually in hand—95.2% are streamable; on the remaining 78.7%, where we fall back to the grounding-free t_{sc} (top-1 stopped moving), 68.1% are. The sufficiency-only number, 95.2%, is the one that matches our framing and is the honest headline for “evidence is retrievable in time”; the 73.9% blend is lower because it is dominated by the t_{sc} majority, for which “streamable” means only that the lexical top-1 has settled, not that gold is in the top-k. Readers who care about evidence sufficiency should weight the 95.2% figure; those who treat top-1 stability as the deployment signal get the blend.

The blend also counts only upside. “Streamable” asks whether enough tool latency *can* be hidden; it is silent on whether the speculative query was *right*, and H’s $\max(0, \cdot)$ floor cannot represent the negative saving a mis-fire incurs (§5.4). One might expect this blind spot to bite hardest in the t_{sc}-fallback majority, which stabilizes *late* (mean ϕ=0.68 vs. ≈ 0.29 on the retrieved-gold replay sample), giving a fixed-interval trigger more room to fire before intent settles. We tested this by replaying 60 fallback questions (the harness previously replayed only retrieved-gold) through the same pipeline at L=600 ms. The expectation does *not* hold: the net-negative-saving rate is 1.7% (1/60), identical to the 1.7% on the favorable retrieved-gold population, and mean savings are likewise comparable. The reason is the same constant that dominates RQ3 (§5.4): the query-generation overlap is hidden regardless of whether the speculative *retrieval* was right, so late stabilization lowers H without proportionally raising the mis-fire rate at this operating point. Mis-fires are thus real but rare ($\approx$ 1.7%) across both populations here; the streamable fraction should still be read as an upper envelope on benefit paired with this small measured downside, and the rate may grow at larger L or with a more aggressive trigger—a sweep we leave to future work.

L-dependent headroom across operationalizations. Table 2 compares the streamable fraction at δ=3 w/s, θ=0.8 across three tool latencies and three corpus arms. Several patterns stand out. First, the streamable fraction declines mechanically with

L for all arms: as the tool call grows, fewer queries have sufficient input residual to cover it. Second, the global fraction is $\leq$ the per-question fraction at every L—the corpus-dependence penalty is always positive. Third, and most importantly, *the gap widens with* L: at L=600 ms, long queries can still hide a modest tool call (58.6% on global NQ vs. 73.9% per-question); but at L=2500 ms—the paper's own calibrated `fuse/reflect` latency of $\approx$ 2520 ms—the global NQ fraction collapses to 9.2% against 39.2% per-question. So the operationalization gap bites hardest precisely in the large-tool-latency regime that motivates streaming RAG.

Three caveats apply. First, the global fractions use the full retrieval corpora (NQ $\sim$1M documents; FiQA $\sim$57k), *not* gold-guaranteed subsamples; gold-guaranteed subsamples would inflate the fraction by construction and are not used here. Second, the two arms use different denominators: the global fraction is over queries whose gold is retrievable at some prefix, whereas the per-question fraction is over the full workload via the t_{sc} fallback. This asymmetry is conservative for our claim—the t_{sc} fallback lets stable-but-unverified top-1 retrievals count as streamable, inflating the per-question fraction, so the reported gap *understates* the true penalty. Third, global sufficiency uses the hard gold-in-top-k criterion, making θ implicit in the global arm; only the streamable threshold $\theta \cdot L$ is shared with the per-question arm.

Turning now to the *input-cadence* sweep (varying δ at fixed L, distinct from the tool-latency sweep in Table 2 above), two further patterns stand out. First, for small tool latency ($L \leq 300$ ms) the constraint is essentially never binding: 82.6% of queries are streamable regardless of cadence—the residual input time trivially covers a short tool call. Second, and confirming **H2**, when tool latency is large (L=1000 ms) the binding constraint becomes residual input time, so a *slower* cadence *raises* the streamable fraction: from 54.5% at δ=4 w/s up to 73.9% at δ=2 w/s. Faster typists and faster talkers leave less room to hide a slow tool.

5.5 RQ3: Does the bound match a working pipeline?

This validation operates under the per-question (C_{perq}) operationalization: the async harness retrieves over each question's own passage pool. Re-plumbing it to a global 10k+ index would require architectural changes to the retrieval back-

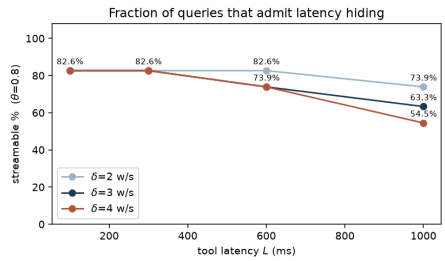

Figure 3: Streamable fraction vs. tool latency L at θ=0.8, one line per cadence δ, with exact values annotated. The curves separate only once L is large enough for residual input time to bind; there, slower cadence dominates (H2).

end that are out of scope here; a global-corpus bound replay is left as future work. The results below therefore validate the H bound *mechanism* under the optimistic operationalization, not under the deployment-realistic global corpus.

We replayed the same 60 retrieved-gold questions through the asynchronous Streaming RAG harness at δ=3 w/s and *two* tool latencies, $L \in \{600, 1000\}$ ms (Fig. 4). In the aggregate the bound is *conservative* at both: measured perceived-latency savings (baseline minus streaming) average 1122 ms vs. the 590 ms H bound at L=600, and 1457 ms vs. 950 ms at L=1000. The overshoot is expected and informative: H models only the tool latency hidden behind input, whereas the calibrated pipeline additionally overlaps the LLM's query-generation step ($\approx$ 590 ms (Arora et al., 2025)) with the user's input. Indeed the overshoot is almost exactly that step: at L=600 the bound mean (590 ms) plus the query-generation constant ($\approx$ 590 ms) recovers the measured mean ($\approx$ 1122 ms) to within noise—a useful sanity check, but also a caution that this paragraph mostly re-derives H plus a constant rather than independently testing H (we address that next). So a query that H deems streamable will, in practice, save *at least* as much as predicted—the property that makes H safe to plan with.

We are deliberately careful about the stronger claim the bound does *not* support: per-query prediction. At L=600 the bound saturates—59 of the 60 queries share H=600 ms (the input residual exceeds L)—leaving no horizontal spread to test ranking, and the rank correlation between H and the measured saving is negligible (Spearman $\rho = +0.14$). We therefore re-ran the *identical* questions at L=1000 specifically to break the saturation:

Table 2: Streamable fraction (percent of queries that can hide $\geq \theta L$ of tool latency) vs. L for the per-question CRAG pool and two global corpora (δ=3 w/s, θ=0.8). The gap between per-question and global fractions widens with L, meaning the operationalization artifact bites hardest in the large-tool-latency regime streaming RAG targets.

Tool latency L	Per-question CRAG	Global NQ $\sim$1M	Global FiQA $\sim$57k
600 ms	73.9%	58.6%	67.6%
1500 ms	54.5%	27.2%	48.8%
2500 ms	39.2%	9.2%	30.8%

H now takes three distinct values (0, 667, 1000 ms) with 7 of 60 queries below the cap. The spread appears, but the correlation does not: $\rho = +0.12$ over all 60, and exactly 0.00 within the unsaturated tail. Two structural reasons explain why raising L cannot rescue per-query prediction. First, the dominant component of realized saving—the constant $\approx$ 590 ms query-generation overlap—is independent of H, so it adds a near-uniform offset that swamps the cross-query signal H encodes. Second, the failure mode is *negative*, and H's $\max(0, \cdot)$ floor cannot represent it. RQ3 thus confirms H as a conservative aggregate bound but *not* a per-query ranker; a predictive model would need to add the query-generation overlap to the hideable budget and replace the floor with a mis-fire penalty.[1]

The negative tail makes the second point concrete and reconnects volatility V to outcomes. The single query that *loses* time—a 17-word comparison ($t^{\star}$=13)—costs -543 ms at L=600 and -940 ms at L=1000, yet its H is *maximal* (1000 ms): with $n - t^{\star}$=4 words of nominal residual the bound rates it fully streamable while the fixed-interval trigger, firing before intent settles at word 13, mis-retrieves and pays a re-fire penalty. Notably this query has V=0, so volatility did *not* flag it—the risk here is *late* stabilization under an early-firing trigger, not post-stabilization churn. This is a single-query observation (n=1 net-negative across the replayed populations), so we offer the reading as a mechanism-level intuition rather than a validated design rule: it *suggests* that an early-commit safeguard should gate on ϕ (how late intent settles relative to the trigger ca-

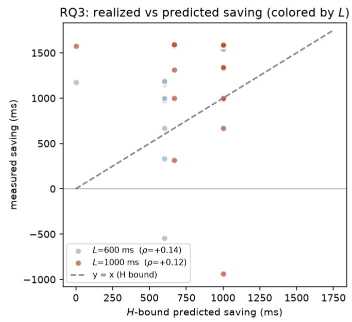

Figure 4: RQ3: per-question measured saving vs. the H bound for the same 60 questions, colored by tool latency L. Most points lie above y=x because the pipeline also hides query-generation latency, which H does not model. At L=600 ms (light) the bound saturates—59/60 points share H=600 ms; at L=1000 ms (dark) H spreads across $\{0, 667, 1000\}$ ms, yet measured savings still do not track it (Spearman $\rho = +0.12$). The lone point below zero is a late-stabilizing query whose mis-fire the floored bound cannot express—and whose H is nonetheless maximal.

dence) at least as much as on V. Whether ϕ in fact out-predicts V for mis-fires requires a larger negative sample—a trigger-aggressiveness and large-L sweep that deliberately induces mis-fires—which we leave to future work and do not claim to have established here.

5.6 Robustness

Grounding (bounding the selection bias). Our sharpest claims live on the 21.3% retrieved-gold slice, whose selection plausibly biases stabilization early (§5.1). To bound that bias we re-derive $d^{\star}$ with a deliberately *relaxed* matcher—bag-of-content-token containment rather than contiguous substring (§4)—which trades precision for recall and so expands the groundable population. Groundability rises from 35.4% to 43.3% and the retrieved-gold rate from 21.3% to 24.4% (a sixth more questions). If the per-question early pattern were largely a selection artifact of the strict matcher, admitting these harder-to-ground questions should pull ϕ_{suf}

[1] Formalizing the first correction without re-running anything: a tighter aggregate bound $H' = \min(L, \text{residual}) + \min(t_{\text{qg}}, \text{residual})$ folds the query-generation step ($t_{\text{qg}} \approx$ 590 ms), which the pipeline overlaps with input regardless of retrieval correctness, into the hideable budget. At L=600 this predicts $\approx 590 + 590 \approx 1180$ ms against the measured 1122 ms—recovering the overshoot to within noise. H' is still aggregate, not per-query: it adds a near-constant offset and so cannot rescue the rank correlation, and it inherits the $\max(0, \cdot)$ floor that the negative tail violates.

up sharply. It does not: mean ϕ_{suf} moves only from 0.26 to 0.281 (median $0.143 \rightarrow 0.154$). The C_{perq} estimate is thus bracketed in $[0.26, 0.281]$—both early within the per-question pool—and the by-type ordering is preserved, now on firmer footing: `set`, the latest type, grows from n=2 to n=20 and remains latest. We read the per-question early-sufficiency pattern as robust to a $1.5\times$ population expansion, not as a grounding-matcher artifact. (Fuzzy grounding is lower-precision, so it if anything biases ϕ_{suf} *down*; that the estimate barely moves is the reassuring direction.)

Grounding (precision). The relaxed arm controls *which* questions are groundable, but not *how specifically* a question's answer is grounded—a distinct bias that, on inspection, dominates the low end of the ϕ_{suf} distribution. CRAG's matcher labels a passage gold whenever the normalized answer string occurs in it, so a short or page-pervading answer (a year, "0"/"1", a yes/no, or a term the whole page is about) is matched by many passages at once; with a gold set that dense, a top-k of size 3 contains a gold passage almost regardless of the query, and t_{suf} collapses to 1 for reasons unrelated to intent. Indeed t_{suf}=1 for 49% of retrieved-gold questions, and their first words are overwhelmingly function words (*what*, *who*, *which*)—which carry no retrieval signal—so the collapse cannot be genuine one-word disambiguation. This precision bias is shared by the exact and relaxed matchers alike and pulls ϕ_{suf} *down*. To bound it we restrict to *specifically*-grounded questions: those whose gold set covers at most 5% of the question's passages, so a random top-3 would hit gold at most $\approx$ 15% of the time. On this subset (n=58) the t_{suf}=1 share falls from 49% to 14% and median ϕ_{suf} rises from 0.14 to 0.35 (mean $0.26 \rightarrow 0.44$); tightening the density cap further moves the estimate monotonically toward $\approx$ 0.4–0.6 rather than back toward zero (the gold-set-density sweep is reproduced by `experiments/grounding_precision.py`). We therefore read the per-question early-sufficiency pattern as *directionally* robust within C_{perq}—sufficiency is still typically reached before the utterance ends in the per-question pool, and well before self-consistency—but materially weaker in magnitude than the uncorrected median 0.14 suggests, and we report the precision-corrected range throughout. The grounding-free ϕ_{sc}, which needs no answer string, is immune to this bias and anchors the comparison.

Table 3: Top-k robustness. ϕ_{sc} (mean 0.67 / median 0.73) is independent of k and omitted.

k	retrieved-gold	mean ϕ_{suf}	median ϕ_{suf}
1	15.6%	0.327	0.200
3	21.3%	0.264	0.143
5	25.0%	0.248	0.125

Retriever (top-k). Varying the sufficiency top-k shifts absolute values but preserves the qualitative picture (Table 3). Larger k surfaces the gold passage for more questions (retrieved-gold rate 15.6% $\rightarrow$ 25.0%) and stabilizes slightly earlier (mean ϕ_{suf} $0.327 \rightarrow 0.248$), as expected since a more permissive top-k is easier to satisfy. The self-consistency curve is unchanged by k (it depends only on the top-1), and the central streamable fraction is stable near 73–74%. Within C_{perq}, the early-sufficiency / late-self-consistency gap holds at every k.

Retriever (dense), per-question pool. A concern specific to the per-question pool is that BM25's reliance on early lexical anchors—not a property of the queries—drives the low ϕ_{suf} we report within C_{perq}. We test this directly by rerunning the full sweep with a dense bi-encoder (`all-MiniLM-L6-v2`, cosine over the same passages) in place of BM25. Under the per-question pool, the dense retriever also stabilizes early: sufficiency reaches the gold passage if anything *later*, not earlier (mean ϕ_{suf} $0.26 \rightarrow 0.31$, median $0.14 \rightarrow 0.22$; ϕ_{sc} $0.67 \rightarrow 0.85$), while surfacing gold for *more* questions (retrieved-gold 21.3% $\rightarrow$ 26.5%). The per-question early pattern is therefore not a BM25 lexical artifact within C_{perq}: a semantic retriever that does not key on early keywords reproduces it at median $\phi_{\text{suf}} = 0.22$. The by-type ordering is preserved (Spearman ρ=0.79 between the BM25 and dense ϕ_{suf} rankings): *aggregation* and *comparison* remain among the earliest under both, and the disagreement is confined to the small-n / grounding-ambiguous cells (`false_premise`, `set`, `post-processing`).

Critically, however, this per-question finding does *not* rescue the early claim under a realistic corpus. The global dense arm—FiQA at gold+10k with the same bi-encoder—yields median $\phi_{\text{suf}} = 0.625$ (see §5.1, Table 1), closely matching the BM25 global figure (0.636). Dense retrieval therefore *generalizes* the corpus-dependence artifact to a non-lexical retriever; it does not rescue the early claim. BM25, if anything, is the *more* optimistic backbone, but the fundamental result—late stabilization under a global corpus—holds for both retriever families.

6 Conclusion

Tool-intent stabilization (TIS) is a query-intrinsic, model-agnostic quantity that bounds when streaming tool use can help and by how much. The central finding of this paper is a measurement caution: *the reported stabilization fraction depends critically on the retrieval corpus*. Against the per-question candidate pool supplied by CRAG—where every document is topically relevant— sufficiency ϕ_{suf} is low (median 0.14 on the retrieved-gold slice, 0.35 after grounding-precision filtering), but that pool is a favorable oracle, not the corpus a deployed system retrieves over. Against a realistic global corpus (NQ at $\sim$1M documents; FiQA at 10k), stabilization is *late*: median ϕ_{suf} reaches 0.75 on NQ and 0.636 on FiQA (BM25), and the same pattern holds for dense retrieval (0.625 on FiQA). These two numbers are not in tension—they measure different operating conditions—but practitioners must report TIS against the corpus their system actually retrieves over.

The TIS framework itself is the durable contribution. It provides a principled, training-free basis for trigger design: ϕ and V characterize per-query stability; $H = \min(L, \max(0, (n - t^\star)/\delta))$ translates stability into a hideable-latency budget that scales with tool latency L. At small L the headroom is modest regardless of corpus (73.9% streamable under per-question, 58.6% under global NQ at L=600 ms); it collapses further at large L (9.2% at L=2500 ms) because later stabilization leaves less residual query to overlap. Question type gives a significant but coarse early/late split (Kruskal–Wallis p=0.017, ε^2=0.06) that is preserved across retrievers (ρ=0.79), providing a practical signal for selective triggering. The system-validation arm confirms H is conservative in aggregate ($\approx$ 1.7% net-negative outcomes at L=600 ms), and the natural next steps—closing the per-query prediction gap and regressing ϕ on linguistic features—build directly on the metrics and data released with this paper.

Limitations

Eight considerations bound how far the present numbers generalize; we take them in turn, from the input model through to the retrieval corpus and benchmark.

Input timing. A uniform word stream is a proxy for speech; real ASR pacing is variable and partial hypotheses get revised. We bound this here by using clean text and treat speech timing as planned follow-up.

Stabilization measure. Top-1 equality is a coarse proxy for "intent settled." We mitigate by also reporting sufficiency against the gold document and by varying top-k.

Grounding. $d^\star$ is derived by string matching, so the groundable population excludes non-verbatim answers; reported ϕ_{suf} statistics are conditional on groundability, and we report t_{sc} as a grounding-free check.

Fallback population. On the 78.7% of queries where BM25 never surfaces the gold passage from any prefix, stabilization is measured only by t_{sc} (when the top-1 stops moving). If the top-1 in this population is the wrong document, early settling is *stable mis-retrieval*, not evidence that the speculative query was right: t_{sc} certifies that intent has *stopped changing*, not that it is *correct*. This is the central gap between our measurement and end-to-end streaming-RAG quality, and it is why we keep the sufficiency and self-consistency populations separate throughout (§5.3) rather than reading the blended streamable fraction as an evidence-quality claim. Our RQ3 replay finds the realized-saving and net-negative rates comparable across the two populations at L=600 (§5.3), but that probes *perceived latency*, not retrieval correctness, on the fallback set.

Per-question candidate pool. The low ϕ_{suf} reported under the CRAG per-question arm (0.14 median) is a property of the evaluation pool, not of the retrieval problem. CRAG supplies a small set of topically relevant passages per question; in that oracle setting the gold document is easy to surface from even a short prefix. This is the source of the v1 over-estimate: restricting to that pool inflates how early stabilization appears to occur. Practitioners should treat the per-question number as an optimistic upper bound, not an operating prediction.

Retriever and corpus. BM25 is lexical and phrasing-sensitive and is known to retrieve on early topical anchors; that behavior could itself drive low ϕ_{suf} in the per-question arm. We addressed this with the dense-retriever arm (§5.6): under the per-question pool, the bi-encoder reproduces the same low-ϕ_{suf} pattern, confirming the per-question result is not a BM25 artifact. However, the global-corpus arm—which is the correct operating condition—shows that *both* BM25 and dense retrieval stabilize late (median ϕ_{suf} 0.636 BM25, 0.625 dense on FiQA at 10k distractors): the oracle artifact *generalizes* to dense retrieval and therefore does not rescue the early-stabilization claim. The by-type ordering is preserved across retrievers (ρ=0.79), so question-type signal remains useful for selective triggering under either retriever.

Global corpus scope. The global-corpus arm uses BEIR-standard constructions: NQ with Wikipedia passage qrels and FiQA with its finance passage pool. These are not the retrieval corpora of any specific deployed system; the true operating ϕ_{suf} is system-specific and will depend on index size, domain, and passage granularity. Our results establish the direction of the corpus-size effect (later stabilization as the index grows), but the exact number must be measured against the target system's own index.

Deferred scale. The full NQ Wikipedia passage index (2.68M passages) and larger dense runs are deferred for compute reasons. The dose-response across 10^3–10^6 distractors and the FiQA dense point at 10k jointly establish the trend, but the asymptotic value under a multi-million-passage index with dense retrieval remains unmeasured. The present results should be read as establishing the direction and magnitude of the corpus-size effect, not its exact operating point at deployment scale.

What remains open is cross-dataset and cross-lingual replication: a single English QA set leaves open whether verb-final word order (e.g., Japanese or German) shifts t_{suf} under lexical but not dense retrieval, which would be a natural next step.

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